

# Machine Learning Overview: Models, Advantages & Disadvantages

*A structured overview of topics, models, and their trade-offs.*

## 1. Data Preparation & Evaluation

| Topic / Section           | Models / Approaches                                                     | Advantages                                                                                                       | Disadvantages                                                                                        |
|---------------------------|-------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| Nominal Feature Encoding  | One-Hot Encoding                                                        | Best choice for nominal variables; avoids imposing an artificial rank; works well with most classical ML models. | Increases dimensionality strongly for high-cardinality features.                                     |
| Compact Category Encoding | Label Encoding                                                          | Compact representation; useful for target labels or truly ordered categories.                                    | Can mislead models into assuming an order where none exists.                                         |
| Missing Value Treatment   | Mean / Median Imputation                                                | Simple, fast, and easy to apply when only a small share of values is missing.                                    | Can distort distributions and weaken relationships between variables.                                |
| Feature Scaling           | Normalization / Standardization                                         | Important for distance-based and gradient-based models; prevents large-scale features from dominating.           | Adds preprocessing complexity and is unnecessary for some tree-based methods.                        |
| Model Evaluation          | Evaluation Metrics (Accuracy, Precision, Recall, F1, ROC-AUC, MSE, MAE) | Allows evaluation to match the task and business goal instead of relying on one score only.                      | Poor metric choice can hide important model weaknesses, especially with imbalanced data or outliers. |

## 2. Unsupervised Learning

| Topic / Section           | Models / Approaches           | Advantages                                                                                             | Disadvantages                                                                                         |
|---------------------------|-------------------------------|--------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Centroid-Based Clustering | K-Means                       | Fast, scalable, and easy to interpret for compact cluster structures.                                  | Requires predefined K; sensitive to initialization and outliers; best for roughly spherical clusters. |
| Density-Based Clustering  | DBSCAN                        | Detects clusters of arbitrary shape and identifies noise points automatically.                         | Sensitive to density parameters; struggles when cluster densities vary strongly.                      |
| Probabilistic Clustering  | Gaussian Mixture Models (GMM) | More flexible than K-Means because clusters can be elliptical; provides soft membership probabilities. | More computationally expensive and based on mixture-distribution assumptions.                         |

| Topic / Section                 | Models / Approaches                | Advantages                                                                                     | Disadvantages                                                                                      |
|---------------------------------|------------------------------------|------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| Linear Dimensionality Reduction | Principal Component Analysis (PCA) | Reduces dimensionality, removes redundancy, and is useful for visualization and preprocessing. | Ignores labels and can make transformed components harder to interpret than the original features. |

### 3. Supervised Learning

| Topic / Section                    | Models / Approaches                  | Advantages                                                                                                                      | Disadvantages                                                                                                        |
|------------------------------------|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| Instance-Based Learning            | K-Nearest Neighbors (KNN)            | Simple, non-parametric, and adaptable for both classification and regression.                                                   | Scales poorly on large datasets; suffers in high-dimensional spaces; sensitive to feature scaling.                   |
| Linear Classification & Regression | Linear / Logistic Regression         | Highly interpretable, statistically grounded, and fast to train; strong baseline models.                                        | Limited to linear relationships unless features are engineered; can be sensitive to outliers.                        |
| Rule-Based Prediction              | Decision Trees                       | Easy to explain; handle mixed feature types; require little preprocessing.                                                      | High variance and strong tendency to overfit if complexity is not controlled.                                        |
| Margin-Based Classification        | Support Vector Machines (SVM)        | Strong in high-dimensional spaces; margin maximization supports good generalization; global optimum for the training objective. | Computationally expensive on large datasets; kernel and hyperparameter choice is critical; limited interpretability. |
| Overfitting Control                | L1 / L2 / Elastic Net Regularization | Reduces overfitting; L1 can perform feature selection; Elastic Net works well with correlated predictors.                       | Requires tuning of penalty strength; too much regularization can lead to underfitting.                               |

### 4. Ensemble Methods

| Topic / Section          | Models / Approaches | Advantages                                                                                                         | Disadvantages                                                                          |
|--------------------------|---------------------|--------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| Bagging-Based Ensembles  | Random Forest       | Reduces variance through bagging and feature randomness; robust on noisy data; usually strong default performance. | Less interpretable than a single tree; slower at inference than simpler linear models. |
| Boosting-Based Ensembles | Gradient Boosting   | Often delivers very high predictive accuracy by sequentially correcting earlier errors.                            | More sensitive to hyperparameters and can overfit if boosting is too aggressive.       |

### 5. Neural Networks & Deep Learning

| Topic / Section             | Models / Approaches            | Advantages                                                                    | Disadvantages                                                                               |
|-----------------------------|--------------------------------|-------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Gradient-Based Optimization | Batch GD / SGD / Mini-Batch GD | SGD and mini-batch methods scale to large datasets; batch updates are stable. | SGD is noisy and learning-rate sensitive; full-batch training can be slow and memory-heavy. |

| Topic / Section               | Models / Approaches  | Advantages                                                                                                   | Disadvantages                                                                                  |
|-------------------------------|----------------------|--------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| Adaptive Optimization         | Adam / AdamW         | Adaptive learning rates often speed up convergence; AdamW usually improves regularization behavior.          | Requires more memory and still needs tuning; not always optimal for every dataset.             |
| Dense Feedforward Networks    | MLP (Dense Networks) | Universal function approximators that work well on many tabular and generic nonlinear tasks.                 | Hard to interpret and inefficient for strongly structured data such as images.                 |
| Convolutional Architectures   | CNN                  | Exploit local structure and weight sharing; highly effective for image tasks.                                | Require substantial compute and are less natural for non-grid data.                            |
| Sequential Memory Models      | RNN                  | Designed for sequential data such as time series or language.                                                | Suffers from vanishing and exploding gradients on long sequences.                              |
| Long-Context Sequence Models  | LSTM / GRU / Bi-LSTM | Better at capturing longer dependencies than vanilla RNNs; useful for sequential prediction.                 | Slower and more complex to train than simpler recurrent models.                                |
| Attention-Based Architectures | Transformers         | Capture long-range dependencies, process sequences in parallel, and are state of the art for many NLP tasks. | Very high memory and compute demands; attention cost grows quadratically with sequence length. |
| Representation Learning       | Autoencoders         | Useful for representation learning, denoising, and dimensionality reduction without labels.                  | Good reconstruction does not always mean useful task-specific features.                        |

## 6. Recommendation & Similarity

| Topic / Section                  | Models / Approaches          | Advantages                                                                                  | Disadvantages                                                                      |
|----------------------------------|------------------------------|---------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| Attribute-Based Recommendation   | Content-Based Recommendation | Does not require behavior data from other users; works well for recommending similar items. | Limited by available item features and can create narrow recommendation bubbles.   |
| Interaction-Based Recommendation | Collaborative Filtering      | Learns from user-item interactions and can uncover non-obvious recommendations.             | Cold-start problem for new users and items; performance issues in sparse matrices. |

## 7. Class Imbalance & Anomaly Detection

| Topic / Section                 | Models / Approaches | Advantages                                                                                            | Disadvantages                                                                                |
|---------------------------------|---------------------|-------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| Synthetic Minority Oversampling | SMOTE / ADASYN      | Improve minority-class learning by generating synthetic examples instead of simple duplication.       | Can amplify noise or blur class boundaries when classes overlap strongly.                    |
| Tree-Based Anomaly Detection    | Isolation Forest    | Scalable unsupervised anomaly detection with roughly linear complexity; works well on large datasets. | Offers limited explanation for why a point is anomalous; less stable on very small datasets. |

## 8. Distributed, Transfer & Reinforcement Learning

| Topic / Section                         | Models / Approaches    | Advantages                                                                                                            | Disadvantages                                                                                                 |
|-----------------------------------------|------------------------|-----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| Privacy-Preserving Distributed Training | Federated Learning     | Trains across decentralized data sources while keeping raw data local, which supports privacy-sensitive applications. | Communication overhead, coordination complexity, and possible privacy leakage from updates remain challenges. |
| Reuse of Pretrained Models              | Transfer Learning      | Reduces data, time, and compute needs by adapting a pre-trained model to a related task.                              | Performance drops when source and target domains differ too much; transferred bias may remain.                |
| Reward-Driven Sequential Learning       | Reinforcement Learning | Learns through interaction and reward, making it suitable for sequential decision problems without labeled datasets.  | Sample inefficient, highly sensitive to reward design, and often difficult to train reliably.                 |

## 9. Responsible AI & Alignment

| Topic / Section           | Models / Approaches           | Advantages                                                                             | Disadvantages                                                                                                        |
|---------------------------|-------------------------------|----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| Trustworthy AI Governance | Responsible AI / AI Alignment | Improves trust through fairness, transparency, robustness, privacy, and human control. | Difficult to operationalize consistently; trade-offs between safety, performance, privacy, and usability are common. |